



Department of Computer Science & Engineering

LAB MANUAL

22CS025 - Algorithm Design & Implementation-5Sem-2022

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Course Name	22CS025 - Algorithm Design & Implementation-5Sem-2022



Faculty Incharge

Aim: The Subset Sum problem

The subset sum problem is an important problem in computer science.

The challenge is to determine if there is some subset of numbers in an given array that can sum up to some given number S.

For example, if the array is {1, 4, 5, 10, 16} and sum = 9, you should return 1 and if sum = 7, you should return 0.

Input Format

First Line will contain an integer M denoting the sum.

Second Line will contain an integer N denoting the total number of elements.

Third line contains N integers separated by space.

Output Format

Print 1 if some subset found and 0 otherwise.

Sample Input

```
9
5
1 4 5 10 16
```

Sample Output

```
1
```

Solution:

```
class Result
{
    static int subsetSum(int a[], int n, int sum){
        if(sum == 0) return 1;
        if(n == 0) return 0;
        if(a[n-1] > sum){
            return subsetSum(a,n-1,sum);
        }
        int result = 0;
        if(subsetSum(a,n-1,sum) == 1 || subsetSum(a,n-1,sum-a[n-1]) ==1){
            result = 1;
        }
        return result;
    }
}
```

